



Simulation model for analysis and management of the no-show in outpatient clinic

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ABSTRACT

In outpatient management, the lead-time is a critical issue due to its important negative effect on healthcare quality perception. In particular, it generates the phenomenon of "no-show": when patients do not attend their scheduled appointments. In this study, we analyze the process of outpatient booking and its critical issues; in particular, we propose a simulation model to evaluate some different approaches. From our results, the lists cleaning can be considered a good tool to manage and reduce the no-show.

CCS Concepts

• **Computing methodologies**→**Modeling and simulation**→**Simulation types and techniques**→**Discrete-event simulation.**

Keywords

Discrete event simulation; no-show; outpatient; E-health; BPM; healthcare.

1. INTRODUCTION

In many large metropolitan areas, the lead-time for outpatient services is one of the most relevant problems that influences the quality of patient's life and therefore effects the assistance quality [1]. This problem is complex and variable; it depends on several elements: the technological development, the increase in average life expectancy, the relationship between the basic physician and the specialist, the insufficient offer and the ever changing of the health state by the population [2].

The phenomenon can be described, in general terms, as a function of three different perspectives:

- Patients: their behavior is unpredictable, in fact, due to the long lead-time they often make multiple bookings

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without canceling previous appointments. Furthermore, this increases the lead-time itself in a sort of loop.

- Health facilities: access to healthcare bookings should only take place upon the prescription of the GP (General Medical Practitioner). Therefore, it would be avoided (or rather banned) to make multiple bookings through a unique center for outpatient booking and by using appropriate management tools.
- GP: they work with a low rate of accuracy. In Italy, for example, there are distributed approximately 15 outpatient services annually per citizen. Therefore, 900 million of total medical visits are provided [3]. The number of medical visit is so high that it is necessary to adopt appropriate management criteria. In addition, there are still strong inequalities in the application of priority classes by prescribers.

One of the critical aspects that can characterize the outpatient activities is the so-called "no-show"; it consists in the failure to cancel the booking in case the patient fails to appear on the booked day [4]. The patient who does not attend the visit constitutes a loss for the budget of the healthcare company and it promotes unsuitable behavior by medical staff [5].

Several studies [6] used simulation tools, discrete event or agents based, in order to reproduce the behavior of medical staff and patient in the different operative conditions. In such way, it is also possible to analyze quantitatively the process and the corresponding service level. Samorani et al. [7] proved that the "lead-time" and the "no-show" are strictly related; in particular, the number of "no-show" patients is, in some way, proportional to lead-time. So that, the reduction of no-show is a primary target in the management of outpatient services. Given the basic hypothesis that the outpatient offering and therefore the number of slots available for visits cannot increase; the most common method to reduce the impact of no-show is the "overbooking". It consists, given the slot available, in increasing the amount of appointments booked for a patient's medical visit in order to offset the no-show [8]. Giacchetti [9] suggested to determine the percentage of overbooking based on the lead-time but also on environmental and behavioral factors. However, he underlines the risk to increase the waiting time in waiting rooms and therefore to raise the dissatisfaction of the patients. Lee et al. [10] introduce, through a simulation model, a mixed technique to minimize the lead-time and no-show.

In this work, the authors propose a simplified simulation model that reproduces the main components of the process: “lead-time, no-show, overbooking”; moreover, the simulation model introduces a new component called “the list cleaning”. The “list cleaning” consists of an activity carried out by one or more phone operators who must call all the patients on the list in order to confirm their will to perform outpatient care. In this way, it is possible to reorganize the waiting queues in order to eliminate or reduce the spaces left free from renunciations. Moreover, the list cleaning reduces the no-show but requires not a marginal cost to the operator and it could introduce dangerous practices to the user.

The use of simulation is helpful to analyze the different elements of the process but the model has to be shareable with medical operators and in different contexts. The model is realized using the simulation software Simul8.

2. MATERIAL AND METHODS

The first step for realizing the simulation model is to individuate the principal actors and the sequence in which the activities take

place. In particular, the activity diagram was realized and shown in Figure 1.

The bookings of the specialist outpatient clinic by citizen happen with the introduction of the relative request in the waiting lists. The patient requires a booking to the call center operator who communicates the first available date and time so that the patient decides to accept or not (booking drop out) depending on lead-time. Of course, the lead-time depends on the number of available slots offered by the structure. During the waiting time the patient can give up on the visit or the visit can be anticipated due to the list cleaning activity. In the visit day, the patient waits in the waiting room in order to receive the visit. In the diagram, shown in Figure 1, it is supposed a simple mechanism of overbooking that provides the addition of a booking slot for each registered no-show patient. Moreover, in order to make the overbooking effective, the overbooked appointment has a higher priority level.

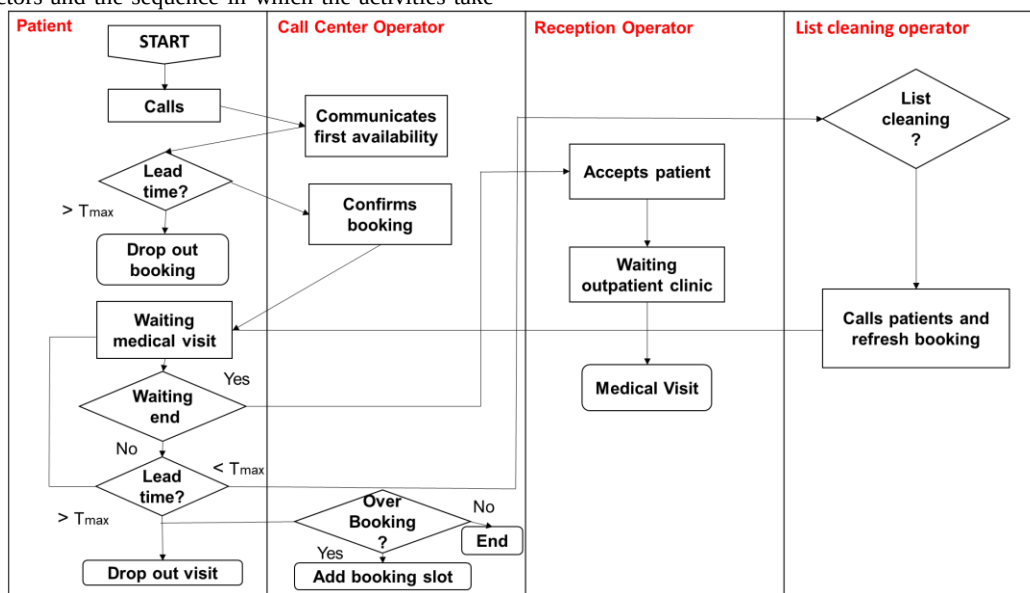


Figure 1. Activity diagram of the outpatient service.

It was realized a discrete event simulation, by using the Simul8 software, to analyze the process dynamics. The model regards the structure with an operator that collects bookings for only one outpatient clinic. Furthermore, the following dynamics are reproduced:

- Booking drop out: the patient calls but he does not make the booking due to the excessive lead-time.
- Visit drop out: the patient of list does not go to the outpatient clinic.
- Over booking: extra booking, over the available slots are made and it is indicated the urgency.
- Cleaning waiting list: a percentage of bookings in list that exceed the max lead-time: T_{wmax} , (one of model parameter listed in the following); these bookings are removed following a telephone recall activity. The latter activity involves all patients booked in list and it requires a specific operator.

The model has the following parameters:

- Tel_in: start point for creation of work item corresponding to user calls.
- Waiting time call center: queue for calling in entry.

- Booking back office: it is an activity for normal booking and overbooking.
- Booking drop out: end of simulation model due to calls that do not give rise to a reservation.
- Pre- list cleaning: fictive activity to handle the list cleanup process.
- List cleaning: extractive activity of pending bookings that exceed the set time limit.
- Call list cleaning: activity where the calling operator calls the patients booked to ask if they want to give up or move their appointment in a free slot.
- Enter (pre_acc): activity that draws reservations from the waiting list queue on the day of the medical visit.
- Waiting room: queue that reproduces the patient waiting in the outpatient clinic.
- Out_door: activity that intercepts and sends to the end the medical visit in excess of the visit time because of excessive booking.
- Visit extra-time: end point to count the number of extra-time visit.

- Outpatient clinic: activity related to the booked visit; it allows recording the medical doctor's commitment and all resources associated with the visit.

The configuration variables indicated above are in a configuration spreadsheet containing the following parameters:

- T_{wmax} (Max Wait Time in day): maximum lead-time threshold beyond which the no-show phenomenon begins.
- I_{cl} (list cleaning): 1/0 whether or not to activate the list cleaning.
- N_p (n_visits a week): number of medical visits a week scheduled in the agenda (loaded at each reset of the model).
- N_{tel} (n_calls a week): number of requests for appointments at week (loaded at each reset of the model).
- N_{ovb} : number of medical visits in overbooking.
- T_p : minimum duration of medical visit (loaded at each reset of the model).
- I_{ovb} (flag ovb): 1/0 whether or not to activate the list cleaning (loaded at each run of model).

It is possible to consider other parameters concerning the economic profit related to both the medical visits and the staff costs in the different operating conditions:

- C_{op} : annual cost of operator (€) (loaded at each reset of model).
- C_m : annual cost of medical doctor (€) (loaded at each reset of model).
- U : profit for visit (€) (loaded at each reset of model).

Some parameters are loaded at each run of simulation model, instead, other parameters are loaded at reset. This depends on whether or not these parameters affect the structure of the model

or its operating modes.

In particular, changing the parameters at each run, it is possible to modify the operating modes during the simulation (stop & go). This mode, provided by almost any simulation system, allows a certain degree of interactivity that is useful to understand the dynamics of the process.

The simulation model provides two diagrams that are uploaded in real time: one shows the trend of patients show (patients who receive the reservation) and no-show (patients who do not show up); the second reports the trend of lead-time. The diagrams allow to have a feedback in real time and to evaluate how the parameter changes affect the model behavior.

2.1 AS-IS Validation Model

The validation of the model requires the comparison between the trends of lead-time with a real case. For this reason, the "lead-time" trend has been compared with parameters collected in the urology department of the University Hospital Company of University of Naples "Federico II".

In the University Hospital Company, a Unique Booking Center is active; it is equipped with various access channels (phone, fax, email, door, pharmacies and volunteer associations). Moreover, it is only dedicated to first-time bookings (first visit or first diagnostic-therapeutic examination) while the bookings of subsequent visits are made directly from the medical doctor who makes the first visit.

The corporate booking center makes use of overbooking and list cleaning on basis of the operative needs. Normally the overbooking is reserved in order to guarantee, in the appropriate cases, the max lead-time indicated by law. In such cases, the front-end operator searches for the first availability offered by the information system and, if it is not congruent with the priority

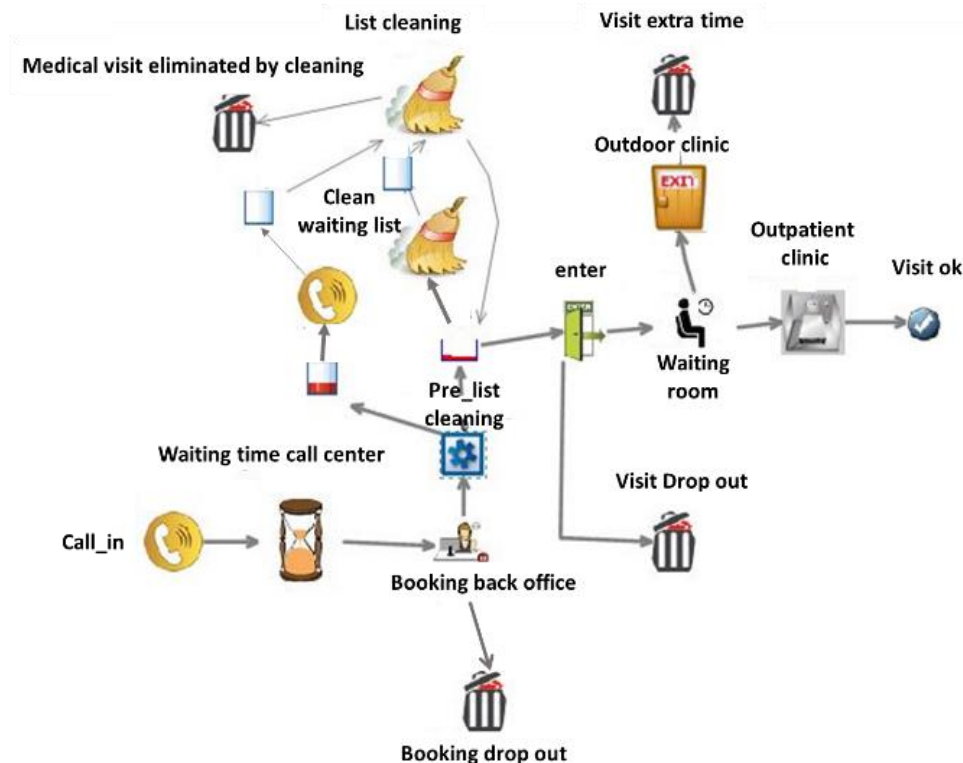


Figure 2. Discrete event simulation model.

given by the prescribing physician on the engagement, makes an extra-appointment (overbooking) in the most congenial session with the urgency. Normally, where it is not specifically required, no solution is adopted to reduce the impact of dropouts.

The benchmarks for the simulation model are:

- Number of week calls to the switchboard: $N_{tel}=200$.
- Number of bookable appointments at week: $N_p=96$.
- Simulation duration: 12 months.
- Max wait time: $T_{wmax}=30$.

All these parameters can be changed in order to adapt the model to specific situation.

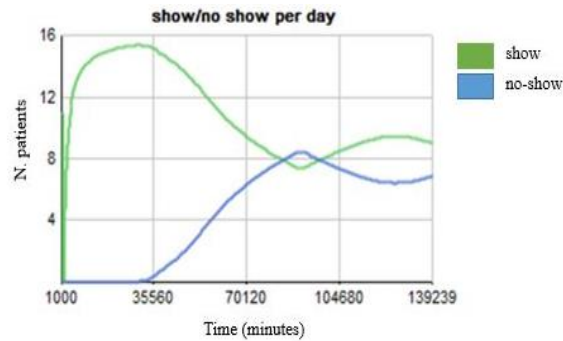


Figure 3. The trend of show/noshow in AS-IS.

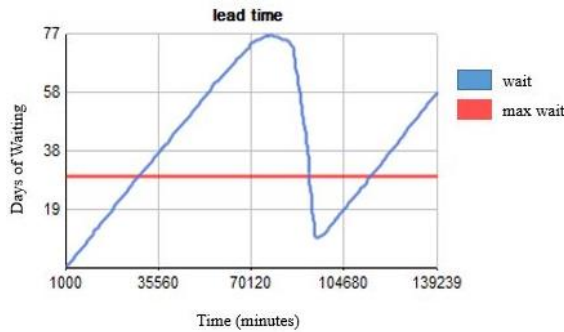


Figure 4. The trend of lead-time in AS-IS.

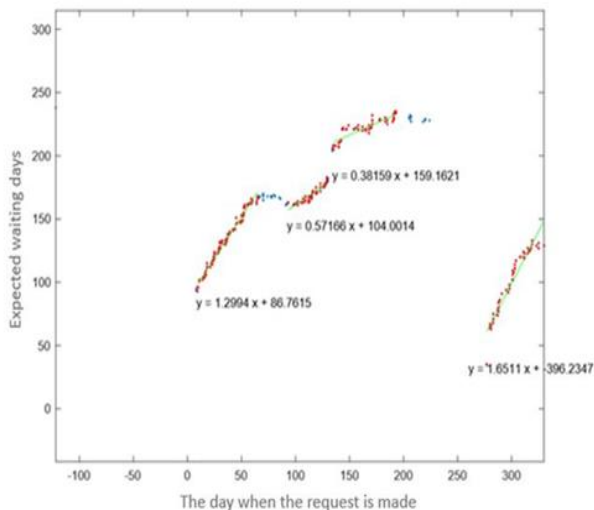


Figure 5. The lead-time trend of real case.

The diagrams (see Figure 3 and Figure 4) report the data after 1-year simulation. The lead-time increases, the no show increases as well and of course after a given time the lead-time decreases and show quota increases proportionally.

In Figure 5, the real case concerning the lead-time trend is reported. The lead-time trend is similar in the two models; the difference in absolute values in lead-time is due to many contextual factors linked also to the behavior of the user. Moreover, from the same data it is noted that the no-show (as reported by the hospital administration) is about 50% (see Figure 3). The validation, of course, is partial and it does not provide the number of medical visits in a quantitatively reliable manner, but, it confirms the proportional link between lead-time and drop out which is the basic general behavior of the process and the simulation model.

2.2 TO BE Model

The principal Key Performance Indicator (KPI) can be synthesized:

- D_{pre} (Drop out a booking): patients who phoned and waived bookings due to the high lead-time.
- D_{post} (Drop out a visit): patients who booked and waived visit due to the high lead-time.
- N_{xtime} : visits extra time due to overbooking or visits out of time and therefore may require additional resources.
- N_{pre} : visits regularly performed.
- U_{pre} : profit of visits.

Based on these indicators, it is also possible to carry out an economic analysis based on the following parameters:

- C_{op} : 90.000€ gross annual cost for a medical doctor.
- C_m : 37.500€ gross annual cost for an operator.
- U : 30€ typical cost for a medical visit.

At the end of the simulation considering that the maximum number of medical visits payable is equal to 4595, we obtain:

- N_{pre} : 2759
- N_{noshow} : $4595 - 2759 = 1836$.

Evaluating the estimated cost of staff obtained and considering the time spent on the visit, we obtain:

Table 1. Cost summary of TO BE model.

Theoretical profit (0 no-show)	€ 51.463,26
Visit gain	€ 82.770,00
Effective cost call center operator	- € 15.625,00
Effective cost of medical doctor	- € 49.091,00
Total profit of the visit	€ 18.054,00

Therefore, in the case indicated, due to the no-show against the theoretical profit shown, there is a proceeds of 18,054€ with a payout of approximately 60% of the expected amount.

3. SIMULATION RESULTS

3.1 Simulation with overbooking

In this model, it is reproduced the overbooking mechanism, adding a number of bookings equal to N_{ovb} which is placed= 2 in the treated case. It is necessary to consider that some medical visits can be delivered out of time compared to the timetable

provided by the visit. This data represent a further KPI to be added and it is called N_{xtime} .

The lead-time trend (see Figure 8) is similar to the previous case because the overbooking does not correct the process but it is possible to notice a sensible decrease of no-show. We have as KPI the following values:

- N_{pre} : 3160
- D_{extra} (Drop out extra time in waiting room): 82
- N_{noshow} : $4595 - 3160 = 1836$.

In this case, the no-show ranges from 40% to 30%.

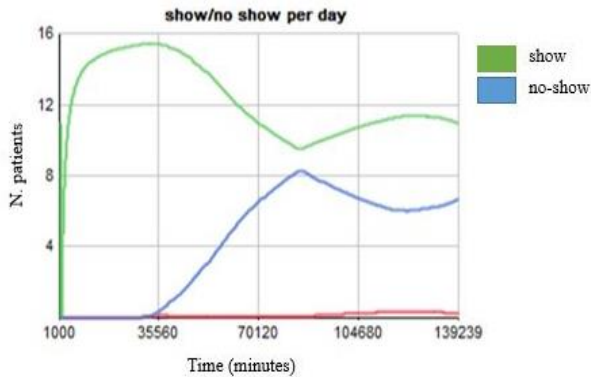


Figure 6. Trend of show/no-show in simulation with overbooking.

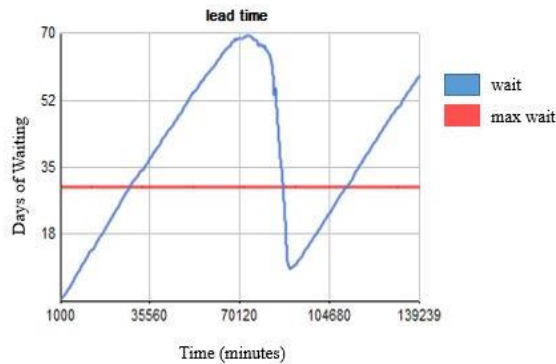


Figure 7. Trend of lead-time in simulation with overbooking.

The results shows a limited economic loss but still high.

Table 2. Cost Summary in simulation with overbooking.

Theoretical profit (0 no-show)	€ 51.463,26
Visit gain	€ 94.800,00
Effective cost of medical doctor	- € 49.091,00
Effective cost call center operator	- € 15.081,00
Total	€ 30.628,00

3.2 Simulation with list cleaning

In the case of list cleaning, assuming we reallocate 50% of the pending appointments (this percentage was obtained by real world experience reported by the medical staff), we obtain the result shown in the following pictures.

In this case, the drop out decreases drastically so that the economical profit increases even though there is a third item cost

associated with the operator for cleaning the lists. The KPI values: $N_{pre} = 4560$.

The results show that when the number of appointment requests is equal to twice the availability, the most economically efficient device to contain the losses due to the no-show is the cleaning of waiting lists.

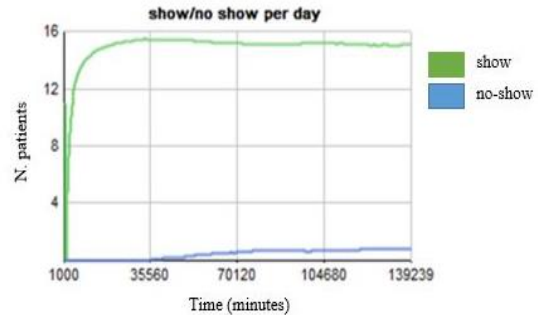


Figure 8. Trend of show/no-show in list cleaning.

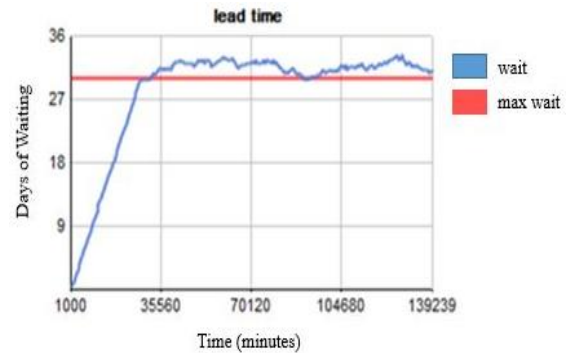


Figure 9. Trend of lead-time in list cleaning.

Table 3 Cost Summary in simulation with list cleaning.

Medical visit profit	€ 138.000,00
Effective cost doctor	- € 49.091,00
Effective cost call center operator	- € 25.212,76
Effective cost of operator of list cleaning	- € 15.625,02
Total	€ 48.071,22

3.3 Simulation outcomes related to number of incoming calls

In order to have a complete picture of the model behavior, one simulation sequence has been done varying the number of incoming calls/day in three different scenarios: normal, with overbooking (for different overbooking levels) and with list cleaning. In particular, the simulations were performed by simulating the following incoming call loads: $N_{tel} = 100, 120, 160, 200, 300$.

For each value, the following KPI were calculated: N. Medical visit; Maximum lead-time (day) and Overall profit.

In case of:

- Normal simulation
- List cleaning simulation
- Simulation with overbooking to following values of overbooking: 2,4,6,8,10.

The diagrams (see Figure 10, 11 and 12), normalized to the maximum value, show that the best strategy is the list cleaning. In fact, the list cleaning allows a reduction of drop out, therefore an increment of number of visits and a leveling of lead-time. The cost of the operator must be amortized on the cost of visits, which are retrieved with the drop out. The revenue (see Figure 11) is lower in list cleaning but from comparison between list cleaning and overbooking it results that the method of list cleaning is more convenient when $N_{tel} > 160$; for lower values is more convenient to use the overbooking technique.

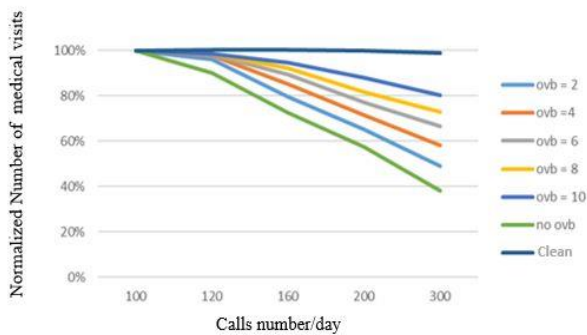


Figure 10. Number of medical visits.

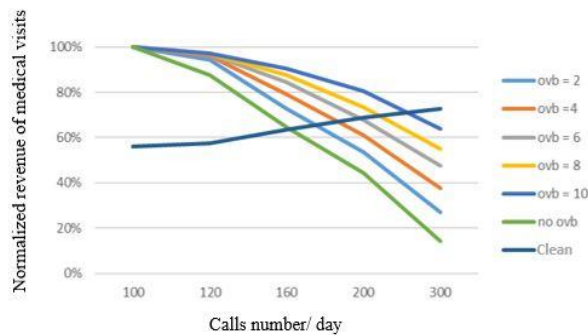


Figure 11. Revenue of medical visits.

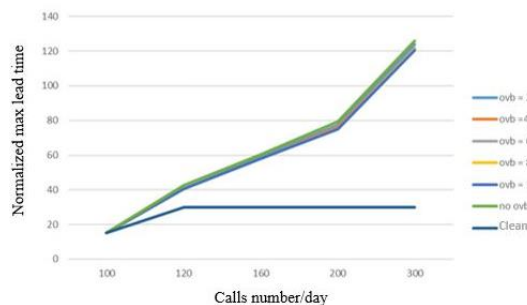


Figure 12. Max lead-time.

These results also depend on the relation between the costs of the medical visit and staff, in particular in the example it refers to:

- Annual cost of medical doctor/ visit cost= 3000.
- Annual cost of operator / visit cost= 1250.

The overbooking becomes more efficient than the list cleaning when the operator cost increases in comparison with the visit cost. However, the list cleaning is necessary for huge lead-time.

4. CONCLUSIONS

The issue of lead-time in outpatient clinic is crucial in many countries and in particular in several areas of Italy. The different national organizations also impose different approaches.

Unfortunately, in the Italian healthcare system, in which most of medical examinations are free or cheap, an increase in the offer often does not solve the problem because there is not appropriate prescription management. Hence the over booking and cleaning of the lists are however indicated to reduce lead-time and no-show in order to improve health service quality perception and to make the system over all more efficient and economical.

The presented simulation model allows you to find the most appropriate solution for the performance offer and the cost of the staff, depending on the number of requests.

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